	Academic Semester	Fall 2026-2027	**Academic level**	1
	Course Code	MA104	**Exam**	Quiz 1 — Model C
	Course Title	Discrete Mathematics		
	Time Allowed	30 Minutes	**Total Points**	10
	Course Instructors	Dr. Salah Moawad , Dr. Tamer Seuodi		

Student Name: _____ **Student ID:** _____ **Section:** _____

*Answer **all** questions. Use T / F in truth tables and 1 / 0 in membership tables.*

Question 1 — Choose the correct answer and give the reason (5 points)

One point each: circle the letter of the correct answer **and** write a short reason in the box.

1. If p is **T** and q is **T**, which proposition is **false**?

A) $p \vee q$ **B)** $p \oplus q$
C) $p \to q$ **D)** $p \leftrightarrow q$

Reason:

2. Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4, 5\}$. Then $B - A =$

A) $\{1\}$ **B)** $\{2, 3\}$
C) $\{4, 5\}$ **D)** $\{1, 4, 5\}$

Reason:

3. $p \oplus q \equiv$?

A) $p \leftrightarrow q$ **B)** $p \vee q$
C) $\neg(p \leftrightarrow q)$ **D)** $p \wedge \neg q$

Reason:

4. Is $(p \leftrightarrow (\neg q))$ a **well-formed formula** (WFF)? If not, name the rule it breaks.

A) WFF **B)** Not a WFF

Reason:

5. Let $A = \{2, 4, 6, 8\}$ and $B = \{4, 8, 10\}$. Then $A - B =$

A) $\{10\}$ **B)** $\{2, 6\}$
C) $\{4, 8\}$ **D)** $\{2, 6, 10\}$

Reason:

Question 2 — Truth table (2.5 points)

(a) Complete the truth table of $((p \vee q) \wedge \neg p) \to q$. [2 pts]

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$	$((p \vee q) \wedge \neg p) \to q$
T	T				
T	F				
F	T				
F	F				

(b) Let P be the proposition in (a). Is $P \equiv p \to q$? Justify using your table. [½ pt]

Question 3 — Membership table (2.5 points)

(a) Complete the membership table for the identity $(A - B) \cup B = A$. [2 pts]

A	B	$A - B$	$(A - B) \cup B$	A
1	1			
1	0			
0	1			
0	0			

(b) Is the identity $(A - B) \cup B = A$ **true** or **false**? Justify using your table. [½ pt]

— End of quiz · Good luck —